

CS4225/CS5425 Big Data Systems for Data Science

Revision

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Exam

- Date: Apr 29, Wednesday
- Time: 9:00am – 11:00am
 - Students are allowed to enter the venue 10 mins before exam starts.
 - Students will not be permitted to enter the venue after 1 hour from the start of the exam.
- Venue: MPSH1-A
- F2F - Hardcopy (pen and paper)
- Open Book Exam
 - Any physical materials are allowed
 - Calculator is allowed
 - Any other electronics devices are NOT allowed

Exam

- Focus is on understanding and application, not facts / memorization
- Question structures (total 50 marks):
 - True / False question with a short explanation / justification
 - Application / Scenario Based Question
 - Give you a practical scenario and let you come out a solution / suggestion
- Example questions
 - **Integrative**: Require you to combine knowledge from different weeks of content
 - **“Application”**: Require you to apply your knowledge of fundamental concepts to reasonably practical scenarios.
 - **“Why not”**: Example, Tommy proposed a solution A to solve problem B. Tell me what is the problem with solution A and how to overcome this problem

Scope of Exam

- **Scope**: the content in the lectures and tutorials
- **Out of scope**:
 - The content marked as “optional” or in the appendix
 - Additional information in the comment box
 - Some notes in the comment box is explaining / clarifying the content in the slides, which is not additional information.
- In the following, I will
 - Have a revision on the **key points** that we learnt after recess week.

Spark I

○ Introduction

- In memory processing and easy to use
- Driver and Executors

○ RDD

- Distributed, Immutable, Lazy Transformations, Action to trigger the computation
- Caching an RDD: when it is expensive to compute and needs to be re-used many times

○ DAGs

- The lineage of an RDD, Within Stage (Narrow Transformation), Across Stage (Wide Transformation)

○ DataFrame: the recommended interface

- filter, sort, join, groupby, and etc.

Spark II

- Spark SQL and Catalyst Optimizer
 - Unifies Spark components and permit various languages
 - Tell Spark what to do and then Spark will generate an optimized plan
- Machine Learning Pipeline
 - Pre-process the Data
 - Build the model using Training Data
 - Evaluate the Model using Testing Data
- Implementing ML Pipeline using Spark MLlib
 - Transformer
 - transform() method: map df1 to df2
 - Estimator
 - fit() method: takes in data and outputs a fitted model (“transformer”)
 - Model training stage: iterative distributed in-memory computation
 - Cache training data in memory across iterations
 - Use broadcast variable to save & broadcast weights iteration by iteration

Stream

- Spark: Structured Streaming (latency of a few seconds)
 - Micro-Batch approach, Incremental Execution
 - Five Steps to define a streaming query
 - Stateless Transformation: filter(), map(), etc.
 - Stateful Transformation / Aggregation
 - Not Based on Time: groupBy().count()
 - Based on Time:
 - Event time vs. Processing time,
 - Tumbling Windows, Overlapping / Sliding Windows
 - Watermark: handling late data
- Flink: Real-time streaming processing (latency of milliseconds)
 - a distributed system for stateful parallel data stream processing
 - Event time processing with watermarks
 - State Management: distributed snapshots using checkpoint barriers

Graph

- Simplified PageRank

- Flow formulation
- Matrix formulation

$$r_j^{(t+1)} = \sum_{i \rightarrow j} \frac{r_i^{(t)}}{d_i} \quad \text{or equivalently} \quad r = Mr$$

- PageRank with Teleports

- Flow Equation: $r_j = \sum_{i \rightarrow j} \beta \frac{r_i}{d_i} + (1 - \beta) \frac{1}{N}$
- Google Matrix
 - $r = A \cdot r$
 - $A = \beta M + (1 - \beta) \left[\frac{1}{N} \right]_{N \times N}$

- Topic Specific PageRank

- A topic-specific set of “relevant” pages (teleport set)

- PageRank Implementation

- Pregel
 - Think like a Vertex, superstep, compute()
 - Partition (Edge Cut) and assign to Workers
- Spark
 - Property Graph and graphframe



study bunnies

chibird